

Eigenvalues and inverse problems

A chromatic lower bound for positive-semidefinite nullity with SAP

Difficulty: challenging **Importance:** interesting to specialist

Status: Solved

Rating rationale (historical): Challenging reflects the need for a stronger general PSD/SAP realization bound than current connectivity estimates supply; implication from Hadwiger does not establish equivalent difficulty. Specialist impact reflects the focus on this particular graph nullity invariant.

Literature-dependent resolution - 2026-09-11

Solved. Hall's **Corollary 3.22**, based on **Theorem 3.20**, gives $\nu(H) \geq \delta(H)$ for every graph H . Induced-subgraph monotonicity and a vertex-critical subgraph of chromatic number $\chi(G)$ then prove the exact target $\nu(G) \geq \chi(G) - 1$. The application note supplies the complete monotonicity and coloring argument, including rank-zero and disconnected cases.

The all-graph theorem is due to **H. Tracy Hall**, *The Delta Theorem*, [arXiv:2601.01211v1](#), submitted 3 January 2026. **George Stepaniants**, Department of Computing and Mathematical Sciences, California Institute of Technology, is the author of the [explanatory application note](#); no new theorem discovery or priority is claimed. [Application PDF](#) · [Standalone TeX](#).

A separate [Codex-agent review](#) checked the full essential proof in Hall's preprint and the exact deduction and returned **PASS**. The source remains a preprint; this is independent automated-agent review, not external human peer review or formal verification. Substantial ChatGPT/Codex assistance and the review's precise limits are disclosed in the [submission record](#).

The earlier status checks below are retained as historical records. The original ID, target, path, historical ratings and attributed partial results remain unchanged. Unsupported prior artifact and graph-atlas claims remain withdrawn.

Problem statement

Let G be a finite simple undirected graph on $n \geq 1$ vertices. Let $\mathcal{S}(G)$ be the real symmetric matrices whose off-diagonal nonzero entries occur exactly at edges of G , with unrestricted diagonal.

A matrix $A \in \mathcal{S}(G)$ has the **strong Arnold property (SAP)** if the only real symmetric matrix X satisfying

$$AX = 0, \quad A \circ X = 0, \quad I_n \circ X = 0$$

is $X = 0$, where $\circ$ is the entrywise product. Define

$$\nu(G) = \max\{\dim \ker A : A \in \mathcal{S}(G), A \succeq 0, A \text{ has SAP}\}.$$

Writing $\chi(G)$ for the minimum number of colors in a proper vertex coloring, does every G satisfy

$$\nu(G) \geq \chi(G) - 1?$$

Relevance and ratings

The conclusion asserts the existence of a positive-semidefinite matrix, with an exact prescribed sparsity pattern, of rank at most $n - \chi(G) + 1$ and with an additional nondegeneracy property. Equivalently it constrains structured Gram matrix construction. This is adjacent to core NLA through matrix realization and inverse eigenvalue theory.

References

- F. Barioli, S. M. Fallat, H. Gupta, and Z. Li, *The weak version of the graph complement conjecture and partial results for the delta conjecture*, Discrete Mathematics 349 (2026), 114861, Conjecture 4.1 and §1 definitions ([primary manuscript](#); [journal](#)).
- F. Barioli, S. M. Fallat, and L. Hogben, *A variant on the graph parameters of Colin de Verdière: Implications to the minimum rank of graphs*, Electronic Journal of Linear Algebra 13 (2005), 387–404, pp.388–390 on SAP and nullity parameters ([primary paper](#)).
- M. Chudnovsky and A. Ovetsky Fradkin, *Hadwiger’s conjecture for quasi-line graphs*, Journal of Graph Theory 59 (2008), 17–33, Theorem 1.1 ([primary manuscript](#); [journal](#)).

Status check

This exact inequality is expressly proposed in the 2026 paper. Its weaker independence-number consequence, Conjecture 4.4, is not separately counted. The source explains that the strong delta conjecture and Hadwiger’s conjecture would each imply the assertion; the original unrestricted symmetric delta conjecture in the companion entry is a different statement. Searches for this paper, “nu”, “chromatic”, “positive semidefinite”, “conjecture”, and 2026 found no resolution. No claim is made that these conjectures are logically independent.

The 2026 counterexamples to a Colin de Verdière conjecture about Laplace eigenvalue multiplicities on surfaces concern a different invariant; they do not refute this finite PSD/SAP inequality.

Audit update — 2026-09-10

Partially resolved: the inequality holds for quasi-line graphs, meaning each vertex neighborhood is the union of two cliques. This follows by combining Chudnovsky–Fradkin Theorem 1.1 with the proved inequality $\nu(G) \geq \eta(G) - 1$ recalled immediately before Conjecture 4.1, where η is the largest clique-minor order. This is an implication of two established results, not a claim that the general conjecture is proved. Both full primary texts, the 2026 publication record, and targeted later searches were checked; no general resolution was located.